

APPENDIX B

 TABLE B.1 Isentropic Flow Table ($\gamma = 1.4$)

M	T_t/T	P_t/P	ρ_t/ρ	A/A^*	MFP
0.1	1.002000	1.007018	1.005008	5.821829	1.998998
0.2	1.008000	1.028281	1.02012	2.963520	3.986079
0.3	1.018000	1.06443	1.045609	2.035065	5.949679
0.4	1.032000	1.116552	1.08193	1.590140	7.878913
0.5	1.050000	1.186213	1.129726	1.339844	9.763859
0.6	1.072000	1.275504	1.189836	1.188200	11.59578
0.7	1.098000	1.387101	1.263298	1.094373	13.36728
0.8	1.128000	1.52434	1.351365	1.038230	15.07237
0.9	1.162000	1.691303	1.45551	1.008863	16.7065
1.0	1.200000	1.892929	1.577441	1.000000	18.26651
1.1	1.242000	2.135136	1.719111	1.007925	19.7505
1.2	1.288000	2.424965	1.882737	1.030440	21.15775
1.3	1.338000	2.770744	2.07081	1.066305	22.48855
1.4	1.392000	3.182272	2.286115	1.114926	23.74404
1.5	1.450000	3.671031	2.531745	1.176167	24.92605
1.6	1.512000	4.250414	2.811121	1.250235	26.03696
1.7	1.578000	4.935993	3.128005	1.337606	27.07956
1.8	1.648000	5.745796	3.486527	1.438982	28.05692
1.9	1.722000	6.700636	3.891194	1.555257	28.97231
2.0	1.800000	7.824449	4.346916	1.687500	29.82908
2.1	1.882000	9.144683	4.859024	1.836944	30.63061
2.2	1.968000	10.69271	5.433288	2.004975	31.38024
2.3	2.058000	12.50428	6.075939	2.193131	32.08125
2.4	2.152000	14.62002	6.793689	2.403100	32.7368
2.5	2.250000	17.08594	7.59375	2.636719	33.34993
2.6	2.352000	19.95403	8.483856	2.895975	33.92351
2.7	2.458000	23.28287	9.472284	3.183011	34.46029
2.8	2.568000	27.1383	10.56787	3.500123	34.96284
2.9	2.682000	31.59408	11.78004	3.849768	35.43356
3.0	2.800000	36.73272	13.11883	4.234568	35.87471
3.1	2.922000	42.64624	14.59488	4.657311	36.28839
3.2	3.048000	49.43703	16.2195	5.120958	36.67656
3.3	3.178000	57.21878	18.00465	5.628647	37.04104
3.4	3.312000	66.11746	19.96300	6.183699	37.38349
3.5	3.45000	76.27230	22.10791	6.789621	37.70549
3.6	3.592000	87.83693	24.45349	7.450111	38.00848
3.7	3.738000	100.9805	27.01458	8.169066	38.29378
3.8	3.888000	115.8889	29.80682	8.950585	38.56263
3.9	4.042000	132.7661	32.84663	9.798973	38.81616
4.0	4.200000	151.8352	36.15124	10.71875	39.05544

TABLE B.2 Isentropic Flow Table ($\gamma = 1.33$)

M	T_t/T	P_t/P	ρ_t/ρ	A/A^*	MFP
0.1	1.00165	1.006667	1.005008	5.866473	1.948722
0.2	1.00660	1.026867	1.020134	2.984731	3.887850
0.3	1.01485	1.061210	1.045682	2.047994	5.808022
0.4	1.02640	1.110733	1.082163	1.598603	7.700335
0.5	1.04125	1.176934	1.130309	1.345408	9.556535
0.6	1.05940	1.261825	1.191076	1.191701	11.369180
0.7	1.08085	1.367996	1.265667	1.096361	13.131770
0.8	1.10560	1.498695	1.355549	1.039139	14.838800
0.9	1.13365	1.657933	1.462473	1.009100	16.485840
1.0	1.16500	1.850604	1.588502	1.000000	18.069450
1.1	1.19965	2.082638	1.736038	1.008192	19.587250
1.2	1.23760	2.361167	1.907859	1.031584	21.037740
1.3	1.27885	2.694731	2.107152	1.069083	22.420300
1.4	1.32340	3.093514	2.33755	1.120283	23.735060
1.5	1.37125	3.569617	2.603185	1.185281	24.982790
1.6	1.42240	4.137367	2.908722	1.264569	26.164780
1.7	1.47685	4.813677	3.259422	1.358968	27.282780
1.8	1.53460	5.618456	3.661186	1.469590	28.338890
1.9	1.59565	6.575067	4.12062	1.597817	29.335450
2.0	1.66000	7.710851	4.645091	1.745290	30.274980
2.1	1.72765	9.057716	5.242796	1.913907	31.160140
2.2	1.79860	10.652800	5.922827	2.105829	31.993620
2.3	1.87285	12.539190	6.695248	2.323484	32.778140
2.4	1.95040	14.766800	7.571165	2.569583	33.516400
2.5	2.03125	17.393210	8.562810	2.847130	34.211040
2.6	2.11540	20.484740	9.683623	3.15944	34.864630
2.7	2.20285	24.117550	10.94834	3.510155	35.479640
2.8	2.29360	28.378900	12.37308	3.903265	36.058460
2.9	2.38765	33.368470	13.97545	4.343125	36.603330
3.0	2.48500	39.199920	15.77461	4.834484	37.116410
3.1	2.58565	46.002460	17.79145	5.382498	37.599730
3.2	2.68960	53.922700	20.04859	5.992764	38.055210
3.3	2.79685	63.126560	22.57059	6.671339	38.484630
3.4	2.90740	73.801420	25.38399	7.424767	38.889690
3.5	3.02125	86.158430	28.51748	8.260109	39.271970
3.6	3.13840	100.435000	32.00197	9.184969	39.632940
3.7	3.25885	116.897400	35.87076	10.20752	39.973990
3.8	3.38260	135.844000	40.15964	11.33655	40.296400
3.9	3.50965	157.608000	44.90705	12.58147	40.601370
4.0	3.64000	182.561300	50.15419	13.95235	40.890010

TABLE B.3 Normal Shock Properties for $\gamma = 1.4$

M_1	M_2	P_2/P_1	ρ_2/ρ_1	P_{t2}/P_{t1}	T_2/T_1	$s_2 - s_1$
1	1	1	1	1	1	0
1.1	0.91177	1.245	1.169082	0.998928	1.064938	0.147349
1.2	0.84217	1.513333	1.341615	0.992798	1.127994	0.927027
1.3	0.785957	1.805	1.515695	0.979374	1.190873	2.635738
1.4	0.739709	2.12	1.689655	0.958194	1.254694	5.372076
1.5	0.701089	2.458333	1.862069	0.929787	1.320216	9.134332
1.6	0.668437	2.82	2.031746	0.8952	1.387969	13.87009
1.7	0.640544	3.205	2.197719	0.855721	1.45833	19.50258
1.8	0.616501	3.613333	2.359223	0.812684	1.531578	25.94507
1.9	0.595616	4.045	2.515679	0.767357	1.607916	33.10892
2	0.57735	4.5	2.666667	0.720874	1.6875	40.90801
2.1	0.561277	4.978333	2.811902	0.674203	1.77045	49.26107
2.2	0.547056	5.48	2.95122	0.628136	1.85686	58.09292
2.3	0.534411	6.005	3.084548	0.583295	1.946801	67.33484
2.4	0.523118	6.553333	3.211896	0.540144	2.040332	76.92463
2.5	0.512989	7.125	3.333333	0.499015	2.1375	86.8064
2.6	0.503871	7.72	3.44898	0.460123	2.238343	96.93014
2.7	0.495634	8.338333	3.558991	0.42359	2.342892	107.2514
2.8	0.488167	8.98	3.663551	0.389464	2.451173	117.7306
2.9	0.48138	9.645	3.762864	0.357733	2.563207	128.3329
3	0.475191	10.33333	3.857143	0.328344	2.679012	139.0276
3.1	0.469534	11.045	3.946612	0.301211	2.798603	149.7874
3.2	0.464349	11.78	4.031496	0.276229	2.921992	160.5887
3.3	0.459586	12.53833	4.11202	0.253276	3.049191	171.4107
3.4	0.4552	13.32	4.188406	0.232226	3.180208	182.235
3.5	0.451154	14.125	4.26087	0.212948	3.315051	193.046
3.6	0.447413	14.95333	4.329621	0.195312	3.453728	203.8297
3.7	0.443948	15.805	4.394864	0.179194	3.596244	214.5743
3.8	0.440732	16.68	4.45679	0.16447	3.742604	225.2695
3.9	0.437742	17.57833	4.515586	0.151027	3.892813	235.9063
4	0.434959	18.5	4.571429	0.138756	4.046875	246.4771

TABLE B.4 Normal Shock Properties for $\gamma = 1.33$

M_1	M_2	P_2/P_1	ρ_2/ρ_1	P_{t2}/P_{t1}	T_2/T_1	$s_2 - s_1$
1	1	1	1	1	1	0
1.1	0.911378	1.239742	1.175051	0.998912	1.055054	0.135695
1.2	0.840904	1.502318	1.355527	0.992645	1.108291	0.920104
1.3	0.783604	1.787725	1.539547	0.97882	1.161202	2.668293
1.4	0.736191	2.095966	1.725404	0.956865	1.214768	5.495886
1.5	0.696397	2.427039	1.911577	0.927246	1.269653	9.414983
1.6	0.662601	2.780944	2.096738	0.891011	1.326319	14.38354
1.7	0.633609	3.157682	2.279751	0.849487	1.3851	20.33198
1.8	0.608523	3.557253	2.459664	0.804087	1.446236	27.17807
1.9	0.586655	3.979657	2.635697	0.756175	1.509907	34.83537
2	0.567466	4.424893	2.807229	0.706992	1.576249	43.21805
2.1	0.550528	4.892961	2.973779	0.657607	1.645368	52.2437
2.2	0.535497	5.383863	3.134994	0.608903	1.717344	61.83473
2.3	0.522097	5.897597	3.290627	0.561579	1.792241	71.91917
2.4	0.510097	6.434163	3.440525	0.51616	1.870111	82.43091
2.5	0.499309	6.993562	3.584615	0.47302	1.950994	93.30964
2.6	0.489575	7.575794	3.722889	0.4324	2.034923	104.5007
2.7	0.480762	8.180858	3.855392	0.394436	2.121927	115.9547
2.8	0.472758	8.808755	3.982211	0.359174	2.212026	127.6274
2.9	0.465467	9.459485	4.10347	0.326596	2.30524	139.4788
3	0.458807	10.13305	4.219316	0.296632	2.401585	151.4735
3.1	0.452709	10.82944	4.329917	0.269176	2.501074	163.5798
3.2	0.447111	11.54867	4.435455	0.244097	2.603717	175.7694
3.3	0.44196	12.29073	4.536121	0.221252	2.709524	188.0173
3.4	0.437211	13.05562	4.632111	0.200487	2.818504	200.3013
3.5	0.432822	13.84335	4.723624	0.181647	2.930662	212.6017
3.6	0.42876	14.65391	4.810859	0.164578	3.046006	224.9013
3.7	0.424993	15.4873	4.894012	0.149132	3.16454	237.1848
3.8	0.421492	16.34352	4.973275	0.135168	3.286269	249.4387
3.9	0.418235	17.22258	5.048837	0.122553	3.411197	261.6515
4.0	0.415198	18.12446	5.120879	0.11116	3.539327	273.8127